

ED302: Curriculum & Assessment

Problem Based Learning 1

Group members:

Joseph Geraghty, 13441542

Shauna Kelly, 13432028

Katie Rogers, 13317526

Cian Mac Liatháin, 11476078

Karen Kerrigan, 13732959

Table of Contents

Introduction.....	3
Sequence of Lessons	7
Assessment of Learning and Assessment for Learning	15
Formative Assessments	17
Group Work.....	20
Questioning.....	25
Four Corners	27
Homework	30
Summative Assessment	33
End of Unit Exam/Worksheet.....	33
Collaboration.....	36
Creativity and Innovation	39
Bibliography	40

Introduction

Project Maths is described by the NCCA as an educational development “designed to provide a bridging framework from the revised primary curriculum into second-level”, with the overall aim of promoting “greater maths literacy across the school population” (NCCA, 2015). Mathematical learning throughout both primary and secondary levels is seen as “cumulative, with work at each level building on and deepening what students have learned at the previous level to foster the overall development of understanding” (NCCA, 2015, p7).

It is evident then, that Junior Certificate Mathematics is a core part of students’ mathematical learning, as it acts as the students’ foundation for Leaving Certificate Mathematics as well as reinforces the topics learnt throughout Primary School.

“The study of Junior Certificate Mathematics encourages the learner to use the numeracy and problem solving skills developed in early childhood education and primary mathematics. The emphasis is on building connected and integrated mathematical understanding” (NCCA, 2015, p7)

In planning a sequence of 15 lessons aimed at a 1st year mixed ability group, we felt that it was important to keep that overall philosophy of Project Maths in mind, and made use of resources provided by the NCCA to help us do that:

- The “Common Introductory Course for Junior Cycle Mathematics” is printed in the Appendix of the Junior Certificate Syllabus (2016). It’s described as “the minimum course to be covered by all learners at the start of the junior cycle” with the goal of laying “the foundation for conceptual understanding which learners can build on subsequently”.
- The “Bridging Framework for Mathematics” explains how the Primary School Curriculum “strand units are continued at Junior Cycle in post-primary school” (NCCA, 2015).

By reading the ‘Common Introductory Course’ in conjunction with the Bridging Framework, we began to devise a series of 15 lesson plans aimed at facilitating “a smooth transition for learners from their mathematics in the primary school” (NCCA, 2015, p33). The Common Introductory Course was a useful resource in determining the mathematical content of our individual lessons,

but it leaves the “order in which topics are introduced” to “the discretion of the teacher”, simply advising that “topics and strands should not be treated in isolation; where appropriate, connections should be made between them.”

Before we could plan “the order in which topics are introduced”, we felt it was important to consider the motivations behind the Project Maths initiative, the teaching methods it recommends, and how that would impact our sequence of lessons in comparison to our personal experience of being taught 1st Year mathematics.

The deductive method of teaching is described by Savic (2012) as the “process of thought from general going to particular. The deductive procedure starts with a rule that is applied to specific cases for the purpose of testing its validity, illustrating or further developing it, or solving the problem to which it applies.” This common method of teaching contrasts with the inductive method, which begins with a particular problem and through the inductive procedure students then “arrive at a fact, principle or generalization”.

In hindsight, we can see that our mathematical education in 1st Year, with the older syllabus, was mainly taught deductively. From our experiences studying the modern syllabus and teaching it during placement, we feel that the Project Maths initiative seems to prefer an inductive method of teaching. We also consulted with multiple different mathematics teachers throughout our placement, who agree that the Project Maths syllabus is more suited to an inductive method of teaching. This view is strengthened further by Cosgrave and all (2012, p7).

“One of the key elements of Project Maths is a greater emphasis on an investigative approach, meaning that students become active participants in developing their mathematical knowledge and skills.

Comparing the inductive and deductive approaches, we can see that each teaching method has its own advantages and disadvantages:

Inductive approach (leads from concrete to abstract, from particular to general)	Deductive approach (leads from general to particular)
+ A natural method of making discoveries	+ Elegant
+ Encourages active participation	+ Puts more emphasis on memory
+ Helps to understand the formulas	+ Time saving
+ Facts are more meaningful	+ Speed and efficiency of doing sums increases
+ Based on observation, thinking and experimentation	- Suitable for mathematical proofs
+ Most suitable in the initial stage of the lesson	- Students are passive learners
+ Learner-centered meaning that it imposes more responsibility on students for their own learning	- Puts more emphasis on memory
- Not absolutely conclusive, hence it cannot be used for exact mathematical proofs but it can be employed for finding mathematical facts	- No discovery element
- Time consuming	- Difficult for a beginner to understand an abstract piece of maths if not preceded by a number of concrete examples

Understanding that Project Maths is intended to be a “bridging framework” between primary and second-level mathematics, and knowing that this sequence of lesson plans is intended for 1st Years of mixed ability, we decided to outline our lessons in a way that would facilitate the inductive approach to teaching. Despite a disadvantage of the inductive approach being that it’s

“time consuming”, we feel the advantages listed above are essential for building a solid “foundation for conceptual understanding which learners can build on subsequently”, as advised by the NCCA (2013, p33).

A deductive approach may become more effective in later lessons, after the students have already built a foundation of basics for them to utilize first.

Our sequence of 15 lessons is designed with guided, inquiry and problem-based learning in mind, which are integral aspects of inductive teaching. (Prince and Felder, 2006) By linking certain topics between Strand 3: Numbers and Strand 4: Algebra, we aim to guide the students through the Common Introductory Course in a way that allows them to naturally discover the links between different strands. This strengthens their foundation of understanding, which can then be further built upon throughout Junior Certificate and Leaving Certificate.

Sequence of Lessons

Lesson	Learning Outcomes	Teaching Strategies	Assessment Strategies
Natural Numbers and The Number Line	All: -Define a natural number -Distinguish between infinite and finite -Represent natural numbers on the number line Most: -Understand set notation that applies to natural numbers.	-Teacher Talk -Questioning -Supervised Student Practise	-Questioning -Exit Slip
Operations	All: -Add and subtract natural numbers Most: -Carry out operations using the number line. Some: -Interpret that addition and subtraction are inverse operations.	-Teacher Talk -Questioning -Supervised Student Practise -Group Work -Student Talk	-Questioning -Homework -Group Work
Skip Counting and the Array Model	All: -Carry out multiplication using	-Teacher Talk -Questioning -Supervised Student	-Questioning -Homework -Exit Slip

	skip counting Most: -Utilise the array model for multiplication and division Some: -Deduce that multiplication reverses division i.e. they are inverse operations	Practise	
Properties of Arithmetic: Commutative, Associative and Distributive Laws	All: -Identify that an operation is commutative if a change in the order of the numbers does not change the result. Most: -Point out that the commutative property applies to addition and multiplication, but not subtraction and division. Some: -Demonstrate how the commutative property can be applied to real	-Teacher Talk -Questioning -Supervised Student Practise	-Questioning -Homework -Four Corners

	like examples.		
Properties of Arithmetic: Commutative, Associative and Distributive Laws	All: -Establish that an operation is associative if a change in grouping does not change the outcome. Most: -Recognise that addition and multiplication are associative operations, unlike subtraction and division. Some: -Understand that the distributive property means that multiplication distributes over addition and subtraction.	-Teacher Talk -Questioning -Supervised Student Practise	-Questioning -Homework -Four Corners -Exit Slip
Equations	All: -Make sense of the concept of equality. Most: -Discover what is meant by an equation	-Teacher Talk -Questioning -Supervised Student Practise -Group Work -Student Talk	-Questioning -Group Work -Exit Slip

	Some: -Apprehend the concept of balance and how it demonstrates equality.		
Introduction to Patterns	All: -Recognise that a pattern is a set of numbers, objects or diagrams that repeat in a predictable manner. Most: -Predict what steps come next in a repeating pattern. Some: -Generalise and explain the pattern or relationship between the terms in words and/or numbers.	-Teacher Talk -Questioning -Supervised Student Practise -Group Work -Student Talk -Guided Discovery	-Questioning -Group Work
Representing Patterns	All: -Represent patterns using tables and diagrams. Most: -Comprehend what is meant by a multiple and how it's applicable to patterns	-Teacher Talk -Questioning -Supervised Student Practise -Guided Discovery	-Questioning -Homework -Exit Slip

	and finding a certain step of a sequence. Some: -Create their own strategies to represent patterns.		
Variables and Constants	All: -Explain what a variable and constant are. Most: -Identify that variables change whereas constants remain the same. Some: -Recognise that variables and constants are features of patterns.	-Teacher Talk -Questioning -Supervised Student Practise	-Questioning -Exit Slip
Coefficients and Terms	All: -Understand the meaning of coefficients and terms. Most: -Recognise that the term is a combination of a variable and a coefficient. Some:	-Teacher Talk -Questioning -Supervised Student Practise	-Questioning -Homework -Four Corners

	-Conclude that expressions are made up of a combination of terms.		
Arithmetic Expressions	All: -Comprehend that expressions are a combination of terms. Most: -Recognise that an expression can determine any term in a sequence. Some: -Derive an expression to represent every term in a sequence.	-Teacher Talk -Questioning -Supervised Student Practise -Guided Discovery	-Questioning -Homework
Arithmetic Expressions	All: -Generate an expression to represent the terms in a sequence. Most: -Solve for the n^{th} term of a given expression. Some: -Explain what each term in an expression represents in relation to a pattern.	-Teacher Talk -Questioning -Supervised Student Practise -Group Work -Student Talk	-Questioning -Homework -Group Work -Exit Slip
Linear Relations	All:	-Teacher Talk	-Questioning

	-Extrapolate that each term is connected to the next term by a certain rule or difference. Most: -Distinguish that if the difference between each term in a sequence is common then it is a linear sequence. Some: -Describe linear sequences in words.	-Questioning -Supervised Student Practise -Guided Discovery -Group Work -Student Talk	-Homework -Group Work
Linear Graphing	All: -Notice that a linear expression is a straight line. Most: -Identify from a given graph that the change in a linear sequence is constant. Some: -Recognise that the first term and the common difference in a sequence will determine the steepness of the	-Teacher Talk -Questioning -Supervised Student Practise -Guided Discovery -Group Work -Student Talk	-Questioning -Homework: Revision -Group Work -Exit Slip

	graph.		
Summative Assessment	All: -Demonstrate their own understanding of the topics covered by attempting to answer questions related to the learning outcomes from each of the previous lessons.		

Assessment of Learning and Assessment for Learning

In order to determine, and hopefully ensure, the level of student learning throughout our lesson plans, we had to devise five formative assessment strategies and one summative assessment strategy that could be utilized throughout the sequence of 15 lessons. The NCCA (2015) advise that widespread changes are having an impact “across the curriculum as the focus in assessment activity begins to move from an emphasis on the assessment of learning to include assessment for”, listing the main features of both types of assessment:

Assessment for Learning	Assessment of Learning
an integral part of the learning process	happens after the learning takes place
information is shared with the learner	information is gathered by the teacher
information is available on the quality of learning	information is usually transformed into marks or grades
comparison with aims and objectives is important	comparison with the performance of others
looks forward to the next stage of learning	looks back on past learning

Assessment for Learning, or formative assessment, is informative feedback to learners while they are still learning the topic. Research shows that this feedback while you learn has more affect on student achievement than any other single factor (Petty, 2009, p480). Black & William (1998) explain, however, that significant learning gains due to formative assessment only occur when the assessment results are used by the teacher to improve the instructional and learning process for the benefit of the students .

Assessment of Learning, or summative assessment, is described as a way for teachers “to summarise and record overall achievement at the end of a course, for promotion and certification” (Pres, 2014).

Even though formative and summative assessments are used for different reasons, (Herman et al, 2006) explain that they should both be used within an “integrated system of assessment, curriculum, and instruction”. To be effective in improving the learning and instructional process, the assessments must be prepared in the context of specific content, teaching methods and educational theories.

In our case, our 15 lesson plans were designed to be integrated within a system that utilizes guided and discovery learning, an inductive method of teaching, and a bridging framework between strands. Our assessment strategies were prepared with this in mind, intending not only to allow teachers to assess what learning outcomes are being achieved, but also to support students throughout their learning.

Formative Assessments

Exit Slip

What is an Exit Slip:

The Exit-Slip method of formative assessment, requires students to respond to questions/prompts handed out by the teacher to each individual student at the end of class. Fisher & Frey (2004) explain that there are 3 categories of exit slips:

Prompts that document learning:

e.g. Write one thing you learnt about natural numbers.
What do you know about linear expressions and graphs?

Prompts that highlight the process of learning:

e.g. I didn't understand...
Write one question you have about today's lesson.

Prompts that can evaluate the effect of other assessment methods

e.g. Did you enjoy working in groups today?

Why we chose it:

The NCCA (2015) advise that being able to “reflect on one’s learning” is a “key element” of the Junior Cycle skills. Through reflecting on their day’s work and thinking of what they learnt or had issues with, students can:

- Become more conscious of the knowledge and skills they developed.
- Identify strengths achieved as well as areas for improvement.
- Gain a better understanding of themselves and how they learn.
- Feel more responsibility for their own learning.

The Exit-Slip method of formative assessment helps students reflect on their learning, and thus improve their ability to realise areas of strength and improvement (Cross, 1998). With our 15 lesson plans designed in a way that links topics and promotes inquisitive learning, reflecting on previous lessons will not only help students reinforce that knowledge but also prepare them for upcoming lessons, as the topics are linked and are based on the previous knowledge of earlier lessons.

This method of assessment doesn't take much time out of the class. With the way the prompts can be phrased, even a quick short response from a student can indicate to the teacher whether learning outcomes are being achieved or whether the student is having difficulty. The teacher can then use this information in future lessons, and in combination with other formative assessment methods like questioning, to support the student in areas they're finding particularly challenging or praise them in areas they understand well.

Advantages:

- Quick, useful feedback from the entire class
- Feedback can easily be integrated into future lessons or assessment methods
- Warns the teacher relatively quickly of any widespread issues with the teaching of particular topics

Potential problems and how to avoid them:

The initial introduction of exit slips and their purpose need to be explained to the students, otherwise they may not feel motivated to complete extra 'work' even if it is only responding to a few prompts.

To ensure that the formative assessment can be used in a way that is most beneficial to both the teacher and the students, it's important for the teacher to organise collected exit slips for each student and have a progressive history of slips from each lesson. This way the teacher gets a long-term view of how each student is progressing through the sequence of lessons.

A potential problem would be students copying each other's responses, while the students are filling out their exit slips it would be best for the teacher to keep a keen eye on whether each student seems like they are answering with their own opinions.

Catering for diversity:

Due to the way the exit slips are phrased, students of all levels interpret the open questions differently depending on their skill level and what they themselves expect to be an adequate answer.

Since teachers will be collecting and storing students' exit slips after each lesson, they can easily see if students of different skill levels are consistently struggling, and can improve the teaching methods accordingly.

How it relates to the topic being taught:

Exit slips are versatile in a way that lets them be linked to any topic being taught, or used to get feedback on the teaching and assessment methods used in that lesson (as highlighted in the examples above).

Group Work

Why we chose it:

Group work is essential in every classroom for certain topics as it hugely benefits student learning. It moves away from a teacher-centred lesson and allows for students to become hands on and be given a chance to express their own opinions and form their own knowledge. It creates a much more enjoyable environment in the classroom bringing an element of fun to the lesson by allowing students a bit more freedom. It is also an excellent assessment tool as the teacher can observe each groups input and progress, and see which students show a greater understanding of the topic, and which students are struggling. According to Petty (2009: p233), “Group work is an adaptable and powerful teaching strategy, with marked advantages; it is an essential feature of almost all modern teaching.” While it is not suitable for all lessons, it is an important teaching strategy and form of assessment which should be implemented in the classroom where possible.

Advantages:

- Group work is not only required in some lessons, but can also make them hugely more enjoyable by including it in the class. However, it is important for the teacher to be “able to direct activities meaningfully” to achieve the learning outcomes of the class. (Petty, 2009: p231) Group activities must “sustain pupils’ attention, interest and motivation.” (Kyriacou, 2007: p26)
- It allows for students to demonstrate, to the teacher, their understanding of the topic and their ability to apply this in the given task. This allows for the teacher to assess a student’s learning but it also “gives the students a chance to use the methods, principles and vocabulary that they are being taught.” (Petty, 2009: 232)
- Group work is particularly beneficial for imaginative learners, who work well in groups and enjoy sharing and integrating ideas with others. (Pres, 2014)
- It allows for the involvement of all students. Those who would normally shy away, avoid raising their hand in the class, or answering out are given a chance to become more involved. This gives the teacher a chance to assess every student, as shy students are more likely to contribute to a smaller group rather than the whole class. Classroom resources such as mini whiteboards can be used for group work where particular students don’t like answering aloud.

- While working alone has its own benefits, group work sets up a support system of peer tutoring for students. Stronger students can act as a guide for students who struggle more with the concept of the lesson. This benefits the students themselves as it reinforces the learning outcomes of the class and avoids embarrassment for any student, but also helps the teacher by preventing students from falling behind. (Petty, 2009, p239)
- Group work aids a teacher in their own assessment as it provides feedback on their teaching. “Good group work hands the responsibility of learning over to the students” and from this, the teacher can identify the students mathematical ability and their understanding of the topic so far. (Petty, 2009, p232) The teacher can then improve the lessons or see what needs to be worked on more.
- As group work allows for conversation and discussion amongst the class, it “improves rapport between students, giving all your classes a more trusting and supportive atmosphere”, creating a positive environment in the classroom. (Petty, 2009, p232) This is particularly good in a first year class as students are new to this type of learning environment and it will help their confidence and give them an opportunity to make new friends. This is hugely beneficial for a teacher as a positive vibe in the classroom will promote learning and allow both student and teacher to enjoy the lesson.
- Lessons based primarily on teacher talk may give the students a sense of being neglected, and so group work allows for them to give their insights and opinions, allowing them to be more involved in the lesson. “In group work, students’ opinions are being valued and accepted; in teacher talk they are being ignored.” (Petty, 2009, p232) This in turn will give the teacher a chance to assess students as individuals even though they are in a group setting.

Potential problem areas and avoiding them:

Problem: While group work allows for the involvement of all students, there is always a chance of a dominant student taking the lead in a task and preventing other students from showing their ability and understanding. This can cause friction if there are two strong-headed students in the class who both try to lead, or it may lead to the other group members sitting back and allowing one student to do all the work. This not only prevents the students from gaining as much from the class as possible, but also prevents the teacher from assessing each individual student.

Solution: Group work may suggest that the teacher does not have much involvement in the lesson, this is not the case. Although the teacher takes a less centred role, their involvement in the lesson is vital to it going well. Each group needs to be continuously observed, directed and monitored. To ensure that all students are putting in equal effort and getting the most out of the class, teachers need to be “demanding feedback” from various students and ensuring that groups do not “go off in the wrong direction.” (Petty, 2009, p233) Teachers may also be strategic about their teaching and group students depending on what suits the lesson, such as random grouping, achievement grouping or a deliberate mix. (Petty, 2009: p243)

Problem: If students do not fully understand what is being asked of them during a group task, this can lead to a lot of complications and delays in the lesson, preventing learning outcomes being achieved. Students may become very confused with what they are supposed to be doing, leading them to become frustrated or disinterested and hugely lower the work rate in the classroom.

Solution: Teachers have a huge responsibility in organising group work as “the task needs to be very clearly stated, and broken down if necessary.” (Petty, 2009, p233) At the beginning of the lesson, the teacher needs to inform the students exactly what is expected of them, and insure throughout the class that the students are steered in the right direction.

Problem: Good student behaviour needs to be maintained during the class which can be difficult during group work as students are given the opportunity to talk with fellow students. Students can become easily distracted in group work tasks and one student can affect the learning of several others in this type of lesson.

Solution: Teachers need to be organised and maintain good classroom management when planning and implementing a lesson involving group work. The size of the groups, how the seating is arranged, the length of the task given and many other aspects affect how well the lesson will go and the teacher needs to assess all these potential problem areas when planning the lesson. Teachers should give students a time frame which they have to work within which keeps students motivated to work harder, but also stops the class from dragging as “group work becomes ineffective when used indiscriminately, or when used too often or for too long.” (Petty,

2009, p233) The classroom set up is also important in preventing as much distraction as possible, and in allowing the teacher to be able to assess and monitor student work throughout the lesson.

Catering for diversity:

- Group work is a good way of allowing teachers to assess the ability of the various students in their class. This is very important, particularly in first year, as teachers need to adapt their class to suit the various types of learners in the room and to cater for everyone's needs, which is not always an easy task. Group work allows the teachers to see who tries to sit back and not be involved in the task, and on the other hand see which students are eager to get involved and show their ability. However, group work benefits both of these types of students as the weaker students are encouraged to become more involved and show their understanding, while the stronger student's learning is reinforced by guiding their fellow students in the given task.
- Group work can also cater for different levels of ability in the classroom as different and more suitable tasks can be assigned to different groups. By affective planning, the teacher can assess the students at a suitable level to their needs, and it is not obvious to class if some students are weaker than others. Every student is challenged but feels equal at the same time, creating a positive attitude in the classroom.
- While some students prefer a teacher-centred learning environment, which has many benefits including how "it can be inspiring" and that "it is a rapid method of presenting material", this is not suitable for all students. (Petty, 2009, p162) There are a huge variety of teaching strategies to cater for different student's learning needs and group work can be used to accommodate students who prefer a hands-on approach to learning. Kinaesthetic learners can particularly benefit from activity-based group work as it moves away from a lecture style lesson and gives the chance for students to get physically involved in their learning.

How it relates to the topic being taught:

- The number line is a fundamental part of natural numbers and in the operations involved with natural numbers. Group work can be used in tasks involving the number line as students can construct their own line or be given one and as a team, work on a number of different questions given by the teacher. This allows all students to demonstrate their learning but also caters for

more visual learners as they can see where the answer is coming from. Students who struggle with the concept also get an opportunity to first observe other students carrying out operations and explaining where they got their answer.

- Patterns can also make use of group work as the teacher can provide students with different resources such as coloured coins, blocks, shapes or other various items that can display what a pattern is. This gives students the opportunity to visualise the idea of a repeating pattern and how to determine what comes next. Group work also allows for students to discuss and defend why they think a certain object comes next in a pattern and to be able to see why they are correct or incorrect without just being told. Later in the chapter, understanding and being able to visualise patterns is fundamental to student's progress and so group work is an extremely important tool in this topic. It also allows teachers to be able to quickly assess student groups by looking in and not having to become too involved, maintaining the students feeling of independence and self-learning.

Questioning

Why we chose it and Advantages:

Questioning is one of our key teaching skills and is included throughout our lesson plans as one of our teaching strategies. Most teachers intuitively use questioning as an assessment strategy because it obtains immediate feedback. It has been used by teachers since the time of Socrates in Ancient Greece. Socrates “taught by asking questions and thus drawing out answers from his pupils” (Changing Works, 2002-2015) which is essentially what we were aiming to do through the teaching strategy of guided discovery i.e. teaching by asking as it is referred to in (Petty, 2009.) Questioning is a good formative assessment strategy because it can be used throughout the lesson and the feedback “regarding how well the learning objectives of a given learning activity are being met” (Unknown, 2015.) is instantaneous. The teacher can use it at the beginning of the lesson to assess prior knowledge, during the lesson to assess the success of their teaching methods and any misconceptions the students have and at the end to recap the lesson. It provides opportunity for all students to demonstrate their learning and get positive feedback for their success. (Petty, 2009).

Potential problem areas and avoiding them:

It would be hard to use questioning with a very large class due to limited time and if a student is not likely to be asked a question for a while they are likely to become distracted. Potentially some students could attempt to monopolise the questioning session particularly the students who are bright and know all the answers or those who are loud or looking for attention. But it is possible to avoid this problem by not accepting their answers to every question and asking students to put their hands up or calling upon the students who are quiet or shy to answer.

Catering for diversity:

Questioning is a good formative assessment strategy for catering for diversity. Teachers can use Bloom’s Taxonomy and ask questions that encompass “the full range of cognitive processes” (State of New South Wales through the Department of Education and Communities, 2011). This allows the teacher to ask weaker students questions of lower cognitive requirement such as

recalling something or a question on comprehension and This way the weaker students get praise and encouragement from answering correctly.

How it relates to the topic being taught:

Questioning works well with all types of mathematics because according to (Petty, 2009) questioning “should encourage all students in the class to think” and in maths students are being challenged to think about and visualise for themselves the scenarios presented especially in relation to patterns and graphing. In our particular topics we may ask a student to define the characteristics of a natural number or to predict what comes next in a pattern. If they get it right we can then ask them to explain why they think that to diagnose whether they really understand the logic behind their answer or whether they just guessed. This way, teachers can quickly discover and address any common misconceptions students may be under.

Four Corners

What is Four Corners:

Four Corners is a "teaching strategy (that) can be easily adapted to any grade level and subject". (Schoenherr, A.B, 2014) It is easily set up within the classroom; using the four corners in the room, the whiteboard and A4 sheets, containing four possible answers to a specific question posed by the teacher. The strategy can be modified to be used for only a short period of time or for an entire class period of 40 minutes.

Why we chose it:

Four Corners is a great way for all students to practise communicating with fellow students, for some of whom English may not be a native language, as well as learning in a co-operative group discussion. It's a step by step process, in which the teacher discusses the expectations of the students; and checks for their understanding throughout the activity e.g. showing a "thumbs up" when they understand the concept or a "thumbs down" when they may not understand or are unsure of the concept.

Advantages:

Four Corners allows students to interact, engage, compare and debate why their own opinion may differ from someone else's, as they are exposed to many different viewpoints. It also allows the teacher to observe and assess the students reasoning, i.e. why they may have a stronger opinion on one answer over the remaining three. The teacher could take note of the students who are progressing, but also the students that may not understand the concept by placing a mark alongside their name (on a roll call sheet). This is a great way to engage with the students. The teacher can present students with misconceptions about the topic and ask whether they may strongly agree, agree, disagree or strongly disagree. This will then result in either a pair or class group discussion. It is used to determine what a student already knows. This allows for the improvement of the teaching and learning of the topic, through reflection of the teaching methods, the student's learning and their understanding. It is easily prepared with little preparation needed on the teachers part (Tsivgadellis, 2015).

Potential problem and avoiding them:

Students may copy their peers by following them in an active four corners lesson that involves the students moving to the four corners of the classroom, without necessarily having a full understanding of the given question. Students may also be embarrassed when speaking in public fearing that they may be wrong, or because they may be shy. Regular use of the four corners strategy ensures all students will become confident with speaking out in class as they will be encouraged to voice their opinions. Doing pair/group discussions before sharing aloud within the class reassures that the student and their partner agree in their opinions.

Catering for diversity:

Four Corners allows each student to express their own opinion. Doing group discussions, one-on-one conversations and teacher feedback, the students can learn from each other. For some learners, it may be easier to learn from their classmates than a lecture style lesson with the teacher standing at the whiteboard. The teacher should construct relevant criticism, feedback and assessment and then take each student aside to discuss these. This includes outlining the positives regarding what they already understand and then draw the students focus to the areas which may require more work. It allows the teacher and student to assess and discover where the student is at in the learning process. When we are teaching, we need to be aware of the different learning styles students may have and make sure that they are not prevented from learning effectively. Four Corners caters for variety of learners due to its flexibility and variety. By having the possible answers in each corner the visual learner would look at information placed on the whiteboard, flashcards and on posters. The kinaesthetic learner would be involved by physically moving around the room and help in the setting up of the activity. The analytic learner would require well structured lessons which require thought and questions outlined in a step-by-step format. (ESL KidStuff, 2013) Each type of learner is being given a chance to show their strong points in learning, but also being challenged to adapt their learning at the same time.

How it relates to the topic being taught:

The Four Corners teaching strategy can be designed to suit the needs of the varying topics being covered in class. An example involving 'Strand 3 Numbers' would be if the teacher used the Four

Corners strategy and posed a question on the natural numbers using symbols or number sets i.e. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{N}$. The teacher would then ask the desired question of the students. The teacher would observe, assess and therefore gain an insight into their understanding. The teacher can also review his/her own teaching and may see flaws where it can be improved. In each corner, the teacher could have a Q&A with the students. At the end of the Q&A strategy, some of the students' opinions may be changed, and with further discussion could bring most of the students' opinion towards the correct answer.

Homework

Why we chose it:

Homework helps the development of the student in the subject and also their construction of knowledge of the topic area. It assists the teacher in the evaluation of the learning, in the amount learned and the understanding of the subject. It creates a good environment for learning, as the student will adapt to learning the topic on his/her own instead of constantly depending on the teachers instruction and help. Finally homework reinforces the topic covered in the learning environment. Homework is not something that, by its own definition, is even done in the classroom, yet it can help teachers elicit evidence of student learning. "Homework should be a risk-free chance to experiment with new skills. Homework should require students to apply what they have learned so they find out what they really do understand" (Trumbull and Farr, 2000).

Advantages:

The advantages of homework include the fact that the teacher can spend more time teaching the topic, while the student actually does the topic in the practical sense at home. It gives the student a sense of achievement on completion of the homework. It also reinforces their prior knowledge of the topic and therefore the teacher can move on to other topic areas swiftly.

Potential problem areas and how to avoid them:

There usually is a tendency to emphasize quantity of work and to neglect its quality in relation to learning, this can be avoided by reviewing the homework you give on a daily basis and by encouraging the student to use their discretion using common sense. Problems also arise when the giving of scores is overemphasized and the giving of useful advice and the learning function are underemphasized. This can be avoided by the teacher if they change their strategies and personally help the students using constructive criticism. "It is important to bear in mind that giving a mark in itself does not help students to understand how they might improve their work. The assessment is not formative in intention and it will do little or nothing for learning. In fact, giving marks or grades cancels out the impact of formative comments" (NCCA 2015). In adopting the approach of comment-only marking the teacher gives feedback in three key areas in the student's learning cycle; what the student has done right, what weaknesses there are in the

student's work and what the next steps should be to improve performance or understanding. This kind of feedback will be most helpful when the learning intentions and criteria for success are clearly understood by the students. Comment-only marking does not preclude the giving of marks or grades to an assignment. The teacher can record the student's marks from each assignment and use them to build up a picture of summative achievement for the student.

Catering for Diversity:

Formative assessment should be used frequently to provide effective feedback, both formal and informal. Homework assignments should consist of a range of activities to develop students' different potentials. Different types of assessment (e.g. portfolio, observation, oral questioning) can be used to help students identify their strengths and weaknesses and for teachers to decide on the appropriate learning and teaching strategies for them. Student's needs and abilities are different due to diversity in background, abilities and interests. Different learning and teaching strategies are therefore required to cater for these differences. Us as teachers should take learner diversity into consideration and take appropriate action to help different learners to learn. This can be achieved through effective curriculum planning and by adopting a range of learning and teaching strategies and assessment modes. Homework assignments should consist of a range of activities to develop students' different potentials. "Quality homework can have particularly positive impacts on student learning. The effectiveness of the homework is particularly dependent upon the teacher's ability to construct, resource, scaffold and provide feedback upon appropriate homework tasks that support in-class learning for diverse students and do not unnecessarily fatigue and frustrate students"(Adrienne Alton-Lee). The All-Most-Some technique can be used for homework as students complete as much as they can. However certain expectations must be reached, so students don't get lazy and risk not reaching their potential. Time limits can also be beneficial especially to dyslexic students. Setting a time will help prevent the students from getting frustrated.

How it relates to the topic being taught:

Maths homework is any task assigned to students for them to complete outside of their maths class, and is created to help students prepare to learn new mathematical concepts, practice ones

that have already been introduced, and explore other maths skills. These out-of-class assignments help to reinforce the lessons a child is introduced during the school day.

Summative Assessment

End of Unit Exam/Worksheet

What is it:

An end of unit exam, with each of the 15 question directly linked to the learning outcomes of each of the 15 lessons taught previous to this.

Why we chose it:

- Johnson & Jenkins (2009) advise that:
 - effective summative assessments include “a clear alignment between assessment, curriculum, and instruction, as well as the use of assessments that are both valid and reliable”.
 - when summative assessments reflect the stated learning outcomes, “a well-designed end of unit test provides teachers with information about individual students (identifying any student who failed to meet objectives), as well as provides an overall indication of classroom instruction.”
- We feel that the type of end of unit exam we’ve devised fits the characteristics advised by Johnson & Jenkins (2009). Integrated with the learning outcomes, as well as the formative assessments building up to it, this end of unit test will help provide feedback to the teacher of each individual student’s performance as well as the class as a whole.
- Gives feedback to the teacher on student progress without using much class time.
- Allows students to assess their own improvement and understanding and the areas they need to work on.

Advantages:

- Allows students to reinforce their learning by revising for a given test.
- Teacher can adapt the assessment depending on the varying student ability and learning types in the class, while still keeping the exam integrated with the learning objectives stated in the lessons.

- Can help determine appropriate route for students to take through a later, differentiated, curriculum. A student's progress in learning different learning objectives in a wide range of mathematical topics, as highlighted by this exam, can be used as an indication of how they might progress through increasingly difficult topics (to estimate a student's potential to progress)
- To identify specific areas of difficulty or strength for a given student, and provide personalised feedback based on the quality of their answers, as this would directly reflect whether they have achieved that particular learning outcome
- To plan future classroom activities, and possibly improve the teaching methods for certain topics if the exam highlights a widespread difficulty throughout the class in answering particular question of the exam – as each question links directly to one of the previous lessons.
- To provide an overall progress report to parents (Pres, 2014)

Potential problem areas:

- Summative end-of-unit type exams can sometimes provide information too late about a student's performance (Popham, 1999)
- The exams themselves put the students in a different atmosphere than they're accustomed to in class – students are disconnected from actual classroom practice during this summative assessment, which may impact their performance despite them understanding the topic (Shepard, 2001)
- These types of exams can suffer from “construct underrepresentation” (Messick, 1989), one assessment usually cannot represent the full content area, so only those areas that are easily measured will be assessed.

Catering for diversity:

- In order to cater for diversity, although each question will be directly linked to the main learning outcomes of each topic, the types of questions will be varied enough to be accessible to all types of learners. For example, in a question regarding patterns, students can be given the option to interpret the pattern using, words, formula or by drawing the next expected shape.
- In combination with the formative assessments (Exit slips and Homework) used throughout the 15 lessons, teachers can compare students performance in each question with how they

performed while learning the topic originally. This way teachers can provide personalised feedback based on the students possible improvement, rather than a straight forward grade. This ensures that the exam caters for students of many different levels, as feedback from it is given in the context of how they previously performed in the formative assessments previously.

How it relates to the topic being taught:

- This summative assessment acts as an accumulation of the topics covered throughout our sequence of 15 lesson plans. Each question directly relates to each lesson plan.
- Our 15 lessons plans were devised based on the Common Introductory Course, which advises that once the course is completed, “teachers can decide which topics to extend or explore to a greater depth, depending on the progress being made by the class group”. This summative assessment can act as a quick reference to see how the class understands each of the possible topics, and teachers will find that information extremely useful when deciding what topic to explore in greater depth.

Collaboration

In order to collaborate as effectively as possible, we decided to create a forum where we could all communicate outside of meetings and discuss ideas as efficiently as possible. We did this by creating a WhatsApp group. We also kept a record of our meetings, attendance and matters discussed. Below is the record of our meeting.

Date	Time, Room, Attendance	
Tuesday 15th September	2-4pm E212 Katie Karen Shauna Cian Joe	Each assigned to our PBL groups (Joe, Cian, Karen, Shauna and Katie) We discussed the roles we would each undertake Chairperson: Karen Scribe: Katie Reader: Joe Timekeeper: Shauna Observer: Cian We organised a plan of what we would have done for our next PBL meeting
Tuesday 22nd September	2-4pm E212 Katie Karen Shauna Cian (late) Joe	Used the maths textbook initially as a brief guide through planning the 15 lessons Then diverted to using the Maths syllabus Katie typed up the learning outcomes of the topics, whilst the rest of the group contributed their ideas We all came up with 5 formative assessments and chose 1 each to research

		and type up for our next PBL meeting Exit Slip- Cian Group Work- Karen Questioning- Shauna Four Corners- Katie Homework- Joe
Thursday 24th September	2-4pm AM107 Katie Karen Shauna Cian Joe	Individually read out the researched information we each found on our formative assessments We researched for different teaching and learning strategies
Tuesday 29th September	2-4pm E212 Katie (late) Karen Shauna Cian Joe (late)	We researched for teaching and learning strategies to use
Thursday 1st October	1-2pm Katie Karen Shauna Joe	We read out each of our formative assessments and discussed them
Tuesday 6th October	12.30-1.30pm AM107	Cian, Karen and Katie started researching the summative assessment

	2-4pm E212 Katie Karen Shauna Cian Joe	Shauna and Joe researched for a link with Philosophy and Maths Methodologies
Wednesday 7th October	4-8pm AM107 Katie Karen Shauna Joe	We designed table for our 15 lessons and the learning outcomes, teaching and assessment strategies
Thursday 8th October	9-11am & 4-5pm AM108 Katie Karen Shauna Joe	Typed up evidence of collaboration Began preparing slides for presentation
Friday 9th October	9-12am Group Study Room 1 Katie Karen Shauna Cian (late) Joe	Reviewed completed document and presentation to be submitted by 5pm today

Creativity and Innovation

In devising our sequence of lesson plans and assessment strategies, we remained conscious throughout the module of ways we'd be able to directly integrate our research for this module into our practical teaching experience.

Each of our formative assessments are able to be used in conjunction with one another, which offers us practical choices of formative assessments that we can do in many situations that we find ourselves in during teaching practice.

After discussing with our co-ordinating maths teachers at our placement schools, we are hoping to continue to improve and develop these assessments and lesson plans to help us not only throughout this years placement but also many years to come.

Bibliography

Adrienne Alton-Lee. (2003). *Quality Teaching for Diverse Students in Schooling: Best Evidence Synthesis Iteration (BES)*. Available:

<https://www.educationcounts.govt.nz/publications/series/2515/5959>. Last accessed 8th October 2015.

Black, P., & William, D. (1998). *Assessment and classroom learning*. *Assessment in Education*, 5(1), 7–74.

Changing Works. (2002-2015). *Socratic Questions*. Available:

http://changingminds.org/techniques/questioning/socratic_questions.htm. Last accessed 6th October 2015.

Cross, K. P. (1998). Classroom research: Implementing the scholarship of teaching. In T. Angelo (Ed.), *Classroom assessment and research: An update on uses, approaches, and research findings* (pp. 5–12).

Diamond, D. (2014). *List of Formative Assessment Strategies*. Available:

http://www.ehow.com/list_6498033_list-formative-assessment-strategies.html. Last accessed 20th Sept 2015.

ESL KidStuff. (2013). *6 different types of ESL learners and how to teach them*. Available:

<http://www.eslkidstuff.com/blog/classroom-management/6-different-types-of-esl-learners-and-how-to-teach-them#sthash.5uuCUVYk.dpbs>. Last accessed 8th Oct 2015.

ESOL. (2009). *Four corners*. Available: <http://esolonline.tki.org.nz/ESOL-Online/Teacher-needs/Pedagogy/ESOL-teaching-strategies/Oral-language/Teaching-approaches-and-strategies/Speaking-Listening/Four-corners>. Last accessed 8th Oct 2015.

Fisher, D., and Frey, N. (2004). *Improving Adolescent Literacy: Strategies at Work*. New Jersey: Pearson Prentice Hall.

Herman, J. L., Osmundson, E., Ayala, C., Schneider, S., & Timms, M. (2006). *The nature and impact of teachers' formative assessment practices*. CSE Technical Report #703.

Kyriacou, C. (2007). *Essential Teaching Skills*. 3rd ed. CHELTENHAM: Nelson Thornes Ltd.

NCCA. (2015). *AfL-Homework*. Available: http://www.ncca.ie/en/Curriculum_and_Assessment/Post-Primary_Education/Junior_Cycle/Assessment_for_Learning_AfL_Homework/AfL_Homework.html. Last accessed 8th October 2015.

Petty, G. (2009). *Teaching Today: A Practical Guide*. 4th ed. CHELTENHAM: Nelson Thornes Ltd.

Popham, W. J. (1999). Where large scale assessment is heading and why it shouldn't. *Educational Measurement: Issues and Practice*, 18(3), 13–17.

Pres, J (2014) *Eliciting Student Learning: Learning Styles*, [Lecture to 2BME1], ED295: Mathematics Methodology & Skills of Teaching. NUI Galway. 22nd November 2014.

Pres, J (2014) *Assessment Strategies* [Lecture to 2BME1], ED295: Mathematics Methodology & Skills of Teaching. NUI Galway.

Savic, A. (2012) Teaching and Learning of Mathematics Available: http://www.ricum.edu.rs/files/3_1/Teacher%20trained%20at%20VISER/Teaching%20And%20Learning%20Of%20Mathematics.pdf. Last accessed 6th October 2015.

Schoenherr, A.B. (2014). *Four Corners Teaching Strategy*. Available: http://www.ehow.com/way_5809507_four-corners-teaching-strategy.html. Last accessed 30th Sept 2015.

State of New South Wales through the Department of Education and Communities. (2011). *When one size doesn't fit all!*. Available: http://www.curriculumsupport.education.nsw.gov.au/secondary/pdhpe/prolearn/reading/pr_007.htm. Last accessed 6th October 2015

Trumbul, E, Farr, B.P (2000). *Grading and Reporting Student Progress in and Age of Standards*. Massachusetts: Christopher-Gordon Pub.

Tsivgadellis, M. (2015). *FOUR CORNERS: A COOPERATIVE LEARNING STRATEGY (POST 4 OF 5)*. Available: <http://www.gottoteach.com/2014/08/four-corners-cooperative-learning.html><http://www.gottoteach.com/2014/08/four-corners-cooperative-learning.html>. Last accessed 8th Oct 2015.

Unknown. (2015). *Diagnostic and Formative Assessment*. Available: <http://serc.carleton.edu/introgeo/assessment/formative.html>. Last accessed 6th October 2015.

PBL 1

Joseph Geraghty, 13441542

Shauna Kelly, 13432028

Katie Rogers, 13317526

Cian MacLiatháin, 11476078

Karen Kerrigan, 13732959

Our 15 Lessons

❖ Natural Numbers

- Definition
- Operations of natural numbers
- Number Line
- Commutative, associative, distributive laws

❖ Patterns

- Representing them on tables
- Variables & constants
- Generating expressions
- Linear patterns
- Linear graphing

Our Formative Assessments

- Exit Slips
- Group Work
- Questioning
- Four Corners
- Homework

Group Work

❖ *“Group work is an adaptable and powerful teaching strategy, with marked advantages; It is an essential feature of all most all modern teaching” (Petty, 2009).*

- Advantages
- Potential Problem areas
- How it relates to the topic

Summative Assessment

- End of Unit Test
- Each question corresponds to the learning outcomes
- Advantages
- Potential Problem areas

Catering for Diversity

- Catering for diversity is *“the underpinning pedagogy, and the desire to provide students with opportunities to achieve as highly as possible, irrespective of their current circumstances, and without the constraint of expectations based on prior attainment”* (Ollerton, 2009, p.7).
- Catering for diversity is vital in a classroom, especially a maths classroom. Students mainly struggle with basic understanding and communicating using Mathematical language, this is where new project maths come in *“To understand mathematics, a student needs to be able to read, solve problems, and communicate using technical language in a specialised context. To properly discuss and explain mathematics content, a teacher is required to use technical language.”* (Freeman & Crawford, 2008, p.12)



Collaboration



Q & A

References

- Freeman, B. and Crawford, L. (2008). *Creating a Middle School Mathematics Curriculum for English-Language Learners*. 1st ed. [ebook] London: Sage & Hammill Institute on Disabilities, p.5. Available at:
<http://www.helpprogram.net/DealerDocs/Sage%20Publications%20article.pdf> [Accessed 9 Oct. 2015].
- Mathematics Syllabus Junior Cert Foundation, Ordinary and Higher Level. (2015). 1st ed. [ebook] NCCA, pp.33-35. Available at:
http://www.ncca.ie/en/Curriculum_and_Assessment/Post-Primary_Education/Project_Maths/Syllabuses_and_Assessment/JC_Maths_English_2013.pdf [Accessed 8 Oct. 2015].
- Ollerton, M. (2009). Inclusive Mathematics Classrooms. *Mathematics Teaching*, 216. 5-7
- Petty, G (2009). *Teaching Today: A Practical Guide*. Cheltenham: Nelson Thornes. p219.